ESG INTELLIGENCE REPORT

AI-Driven Industrial Carbon Footprint & ESG Intelligence Report

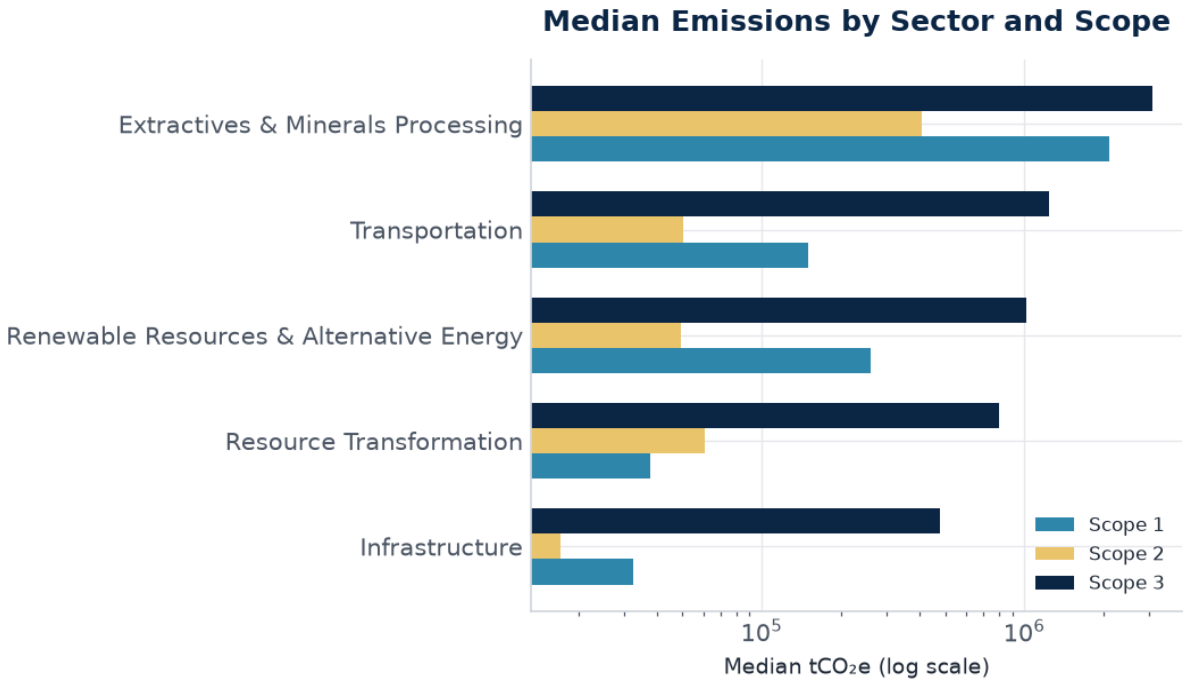
Generated October 19, 2026

Executive Summary

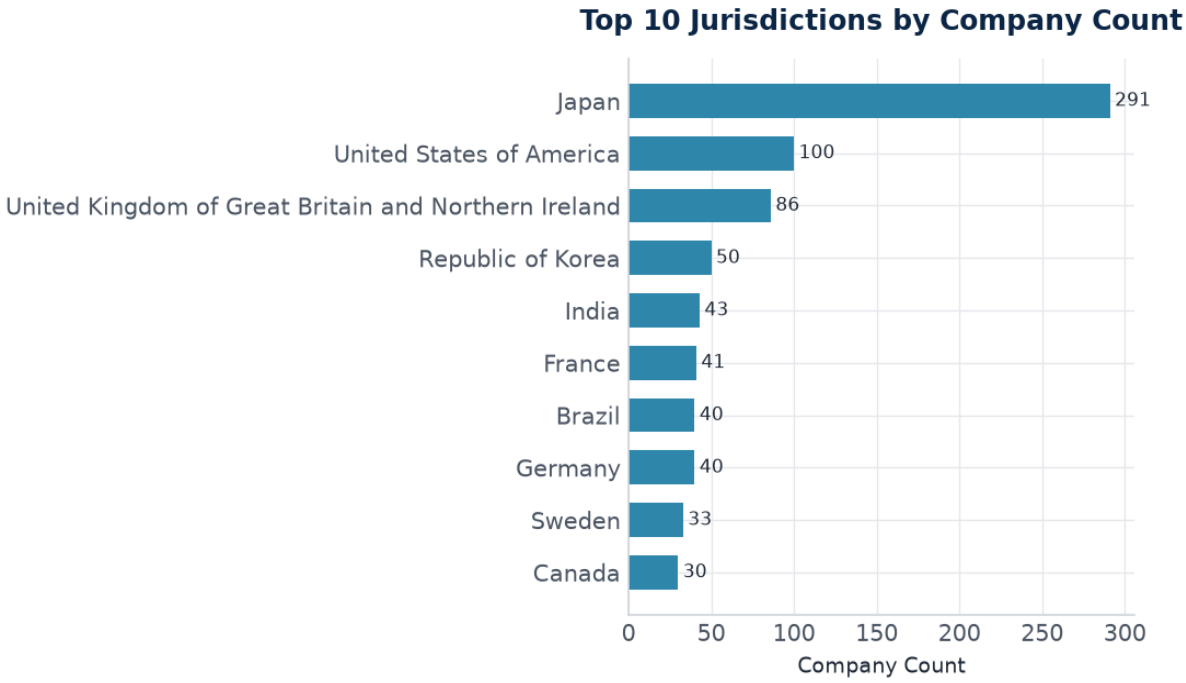
This report summarizes greenhouse gas (GHG) emissions data across 1,069 companies spanning 5 industrial sectors and 51 jurisdictions, sourced from real, verified corporate disclosures (NZDPU / Climate Data Utility, drawing on CDP and ADEME). Aggregate emissions: Scope 1 totals 3,864,059,424 tCO■e, Scope 2 totals 517,150,388 tCO■e, and Scope 3 (value chain) totals 11,927,196,051 tCO■e — consistent with the well-documented pattern that Scope 3 dominates most companies' overall footprint.

Metric	Value
Companies covered	1,069
Sectors covered	5
Jurisdictions covered	51
Total Scope 1 (tCO■e)	3,864,059,424
Total Scope 2 (tCO■e)	517,150,388
Total Scope 3 (tCO■e)	11,927,196,051

Emissions by Sector

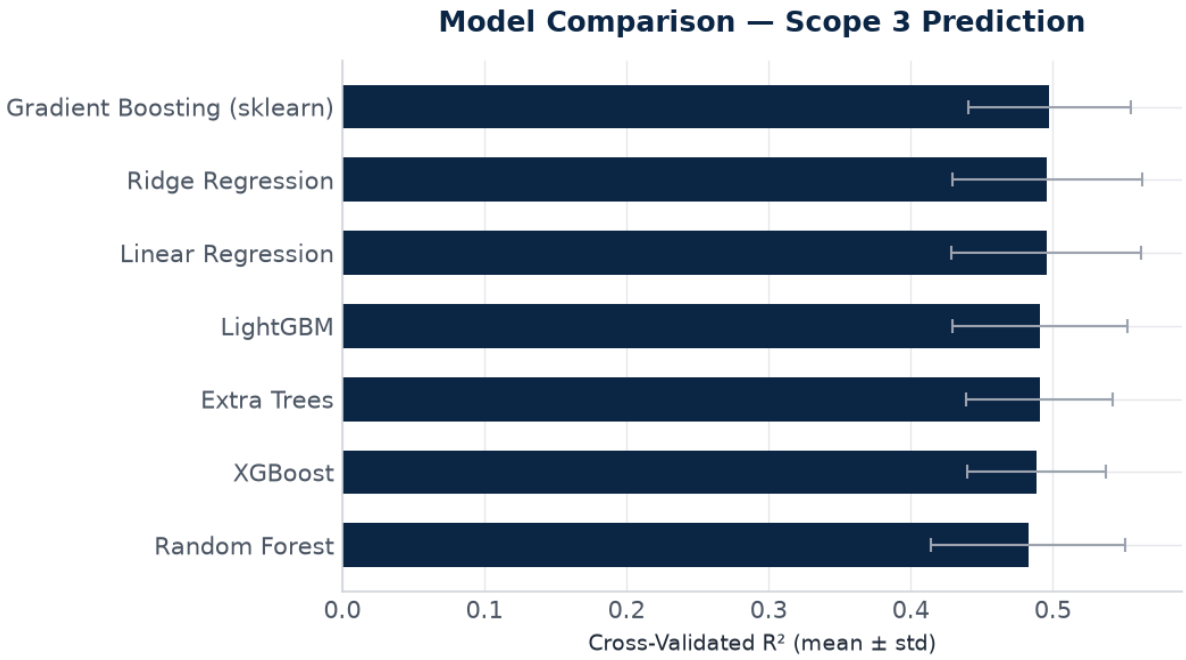


Geographic Coverage



Scope 3 Prediction Methodology

Scope 3 emissions — the hardest to measure of the three GHG Protocol scopes — are estimated using machine learning trained on real company disclosures, sector classification, jurisdiction, and firm financial indicators (revenue, employee count, market capitalization, EBITDA). Financial features were found to be the strongest predictor of Scope 3 emissions, consistent with published benchmark research (GHGbench, 2026). Seven regression approaches were compared under identical conditions.



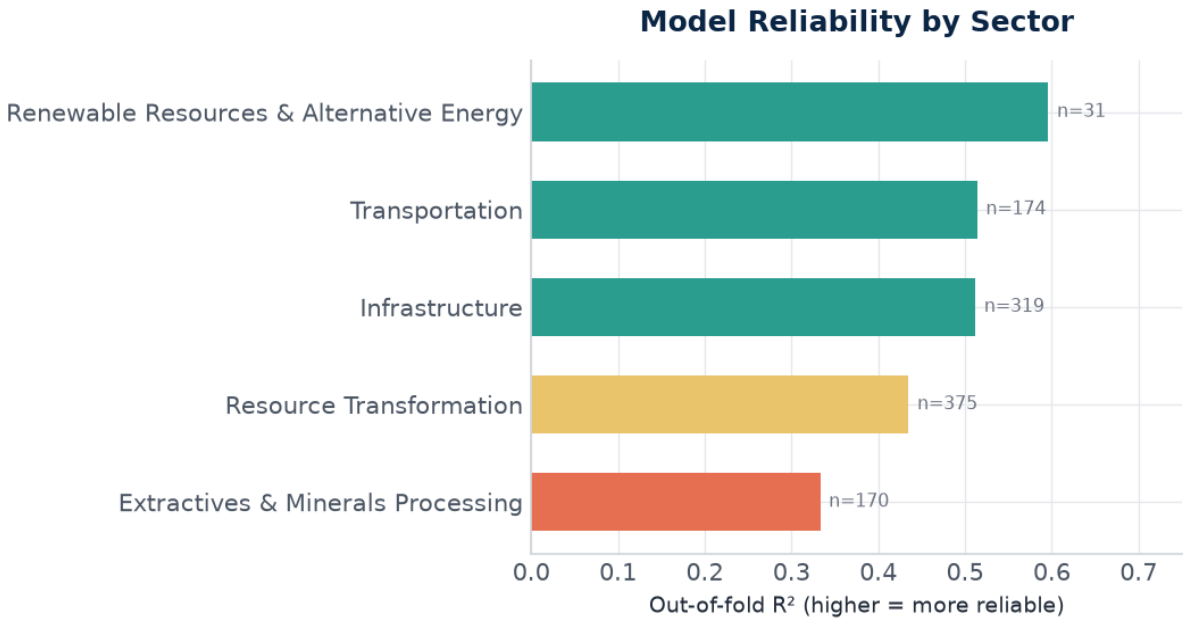
Full Model Comparison

Model	Test R2	Test MAE	CV R2 mean	CV R2 std	Train time (s)
Gradient Boosting (sklearn)	0.4616	1.5883	0.4975	0.0573	1.91
Ridge Regression	0.4676	1.613	0.4959	0.0668	0.07
Linear Regression	0.467	1.6143	0.4956	0.0668	0.08
LightGBM	0.4582	1.6012	0.4911	0.0616	0.42
Extra Trees	0.4476	1.6114	0.4907	0.0517	1.51
XGBoost	0.4628	1.5941	0.4883	0.0488	0.38
Random Forest	0.4367	1.6233	0.4826	0.0684	2.63

R^2 in the 0.48–0.50 range across all seven models is consistent with, and directionally explained by, published Scope 3 prediction benchmarks (GHGbench, arXiv:2605.13743) given this project's smaller sample size relative to that study. XGBoost is used as the deployed model for its lowest cross-validation variance.

Model Reliability by Segment

Out-of-fold R^2 (each company predicted by a model that never saw it during training) broken down by sector, jurisdiction, and company size. This identifies where predictions are most and least reliable.



Segment Type	Segment	R^2	Companies
Company Size (Revenue Quartile)	Mid-Small	0.484	142
Company Size (Revenue Quartile)	Small	0.413	143
Company Size (Revenue Quartile)	Large	0.313	142
Company Size (Revenue Quartile)	Mid-Large	0.29	142
Jurisdiction	France	0.661	41
Jurisdiction	Germany	0.64	40
Jurisdiction	Other	0.527	378
Jurisdiction	Japan	0.513	291
Jurisdiction	United Kingdom of Great Britain and Northern Ireland	0.469	86
Jurisdiction	India	0.413	43
Jurisdiction	Republic of Korea	0.389	50
Jurisdiction	United States of America	0.252	100
Jurisdiction	Brazil	0.193	40
Sector	Renewable Resources & Alternative Energy	0.596	31
Sector	Transportation	0.514	174
Sector	Infrastructure	0.512	319
Sector	Resource Transformation	0.435	375

Sector	Extractives & Minerals Processing	0.333	170

Data Sources & Limitations

Scope 1/2/3 data: Real, verified corporate disclosures sourced from NZDPU / Climate Data Utility (CDP and ADEME providers).

Data quality correction: 172 of 1,206 companies (14.3%) showed Scope 3-to-Scope 1/2 ratios inconsistent with sector norms. 35 were corrected for a diagnosed kg-vs-tonne unit error; 137 were excluded as irreconcilable rather than estimated.

Logistics/routing data: Synthetic, explicitly disclosed. Cost-per-km figures are documented modeling assumptions, not real company pricing.

Model accuracy: $R^2 \approx 0.48–0.50$ on held-out data — a directional estimate, not an audit-grade figure. Weaker for the largest companies by revenue (known tree-model extrapolation limitation).

Regulatory status: AI-generated emission estimates are not universally recognized as valid compliance data by all regulatory bodies.